ViT-GNN: Vision Transformers Meet Graph Neural Networks

Grace Kim· Hideo Tanaka· Frank Schmidt·

Abstract

We present a novel approach combining attention mechanisms with graph neural networks for improved molecular property prediction. Experiments on standard benchmarks demonstrate state-of-the-art performance across multiple datasets.

1 Introduction

Deep learning has transformed many areas of scientific computing. In this paper we propose a unified framework that addresses key limitations of prior work. Our contributions are: (1) a novel architecture, (2) improved training procedure, (3) comprehensive benchmarks.

· Grace Kim · Caltech, Pasadena CA 91125; gkim@caltech.edu
· Hideo Tanaka · Kyoto University, Japan; hideo.tanaka@kyoto-u.ac.jp
· Frank Schmidt · Technical University of Munich; fschmidt@tum.de
Work done while Grace Kim was an intern at Google Research.
Hideo Tanaka acknowledges support from JSPS grant 22K21789.

Acknowledgements

This work was supported in part by a Google Research Award. Grace Kim is a Google PhD Fellow. Hideo Tanaka is supported by JSPS grant 22K21789.